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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-51

DC Charging (CCS2)-II

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Department of Electrical Engineering



Recap: DC Charging System

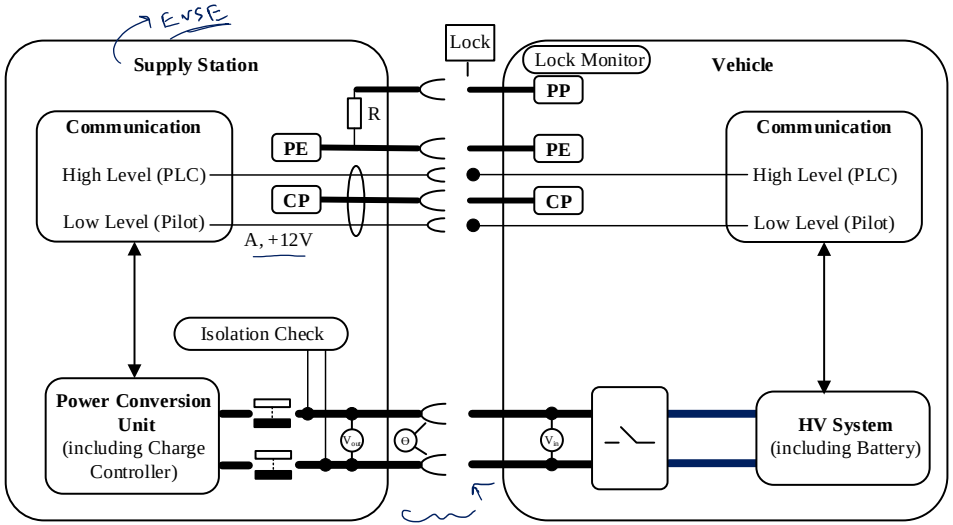
CCS2 → PLC over CP - PE → HLC

1kHz

having

Pwm 3-7%

DC CCS2 Charging Sequence



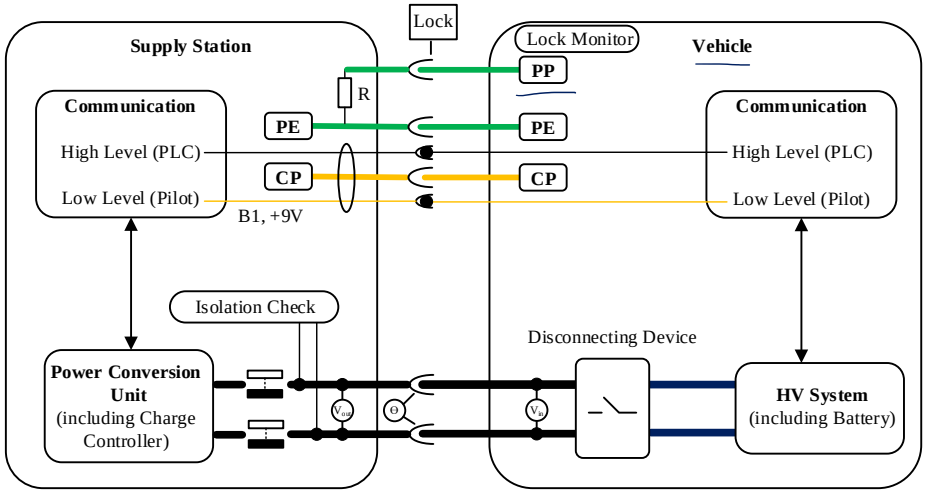
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

- Charger plug is Unmated
- Charging is in State-A

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.

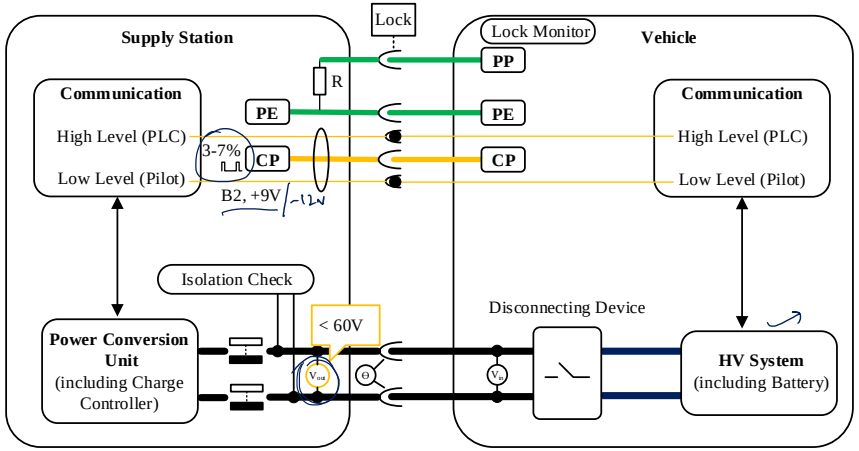
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2



- Charger plug is mated at t_0
- Charging enters state B1 instantly with mating
- Vehicle is immobilized (after detection in PP)

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DC CCS2 Charging Sequence Cont.



Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

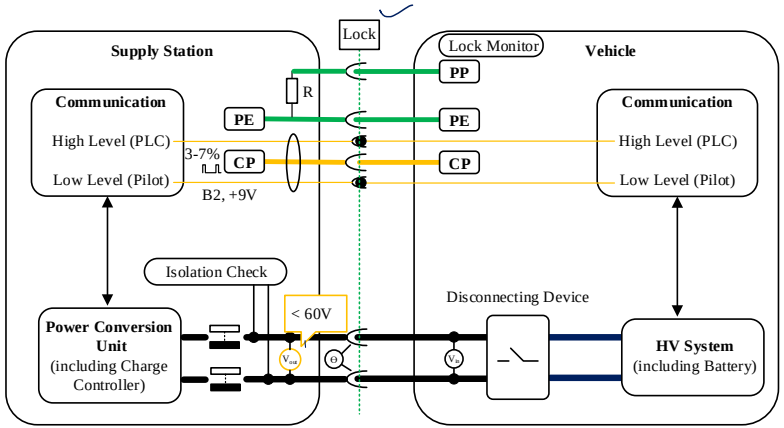
Initializing stage between $t_0 - t_1$

EV sends its maximum limits (amongst other parameters) for DC supply output current and voltage

- High level communication (PLC) starts and handshaking with exchange of charging parameter takes place.
- Shutdown if DC Voltage > 60V or incompatibility of EV and DC supply is detected

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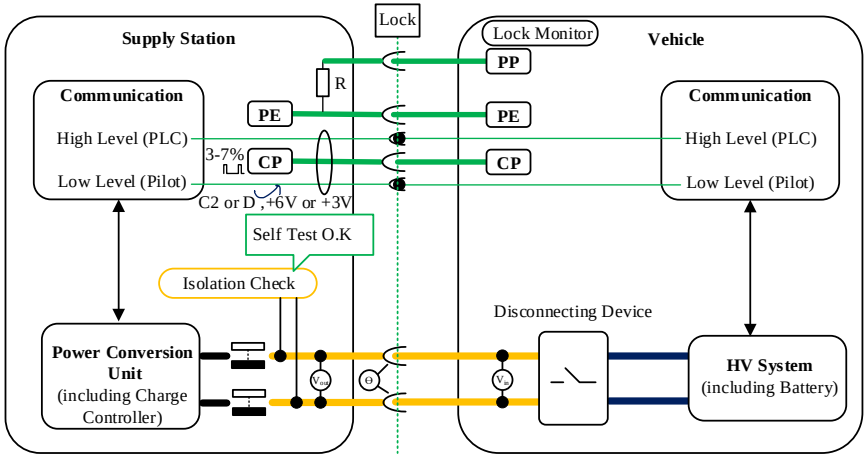
Stage between $t_1 - t_2$

- If EV and DC supply are not compatible, then the vehicle will not go to Ready, and will transition to normal shutdown sequence.

- EV locks vehicle connector in its inlet.
- Maximum values of the DC supply are responded to the EV
- DC supply can check internal insulation as long as no voltage is applied to the connector.

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DC CCS2 Charging Sequence Cont.



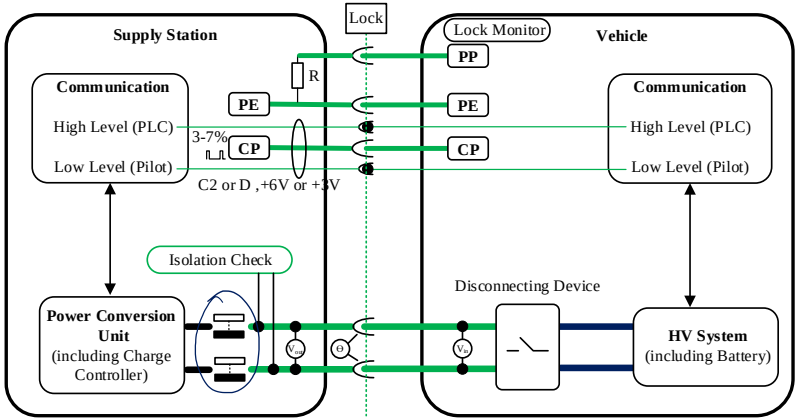
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

Cable check stage between $t_2 - t_3$

- EV changes CP state from B to C2/D and sets EV status “Ready”
- EV requests the cable and insulation check after connector lock has been confirmed
- DC supply starts checking HV system isolation and continuously reports isolation state

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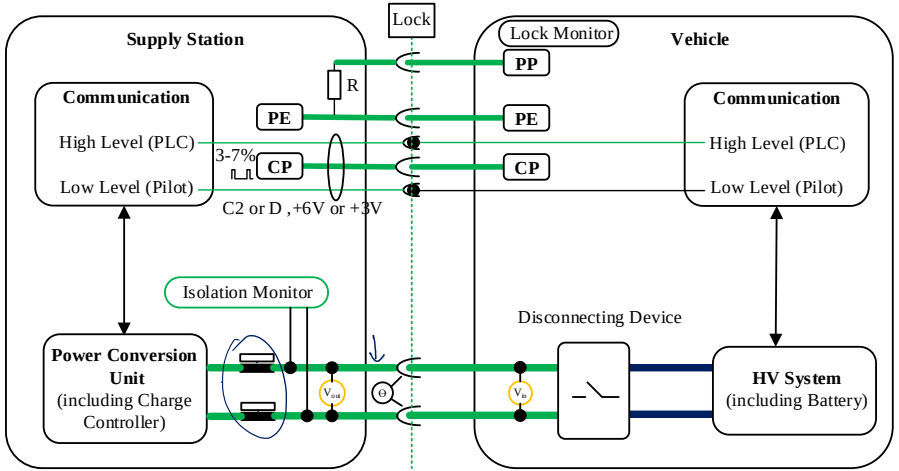
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Cable check stage between t_3 - t_4

- D.C. supply determines that isolation resistance of system is above 100 kΩ.
- After successful isolation check, DC supply indicates status ”Valid” and changes status to “Ready” with Cable Check Response.

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DC CCS2 Charging Sequence Cont.



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Pre-charge at t_5

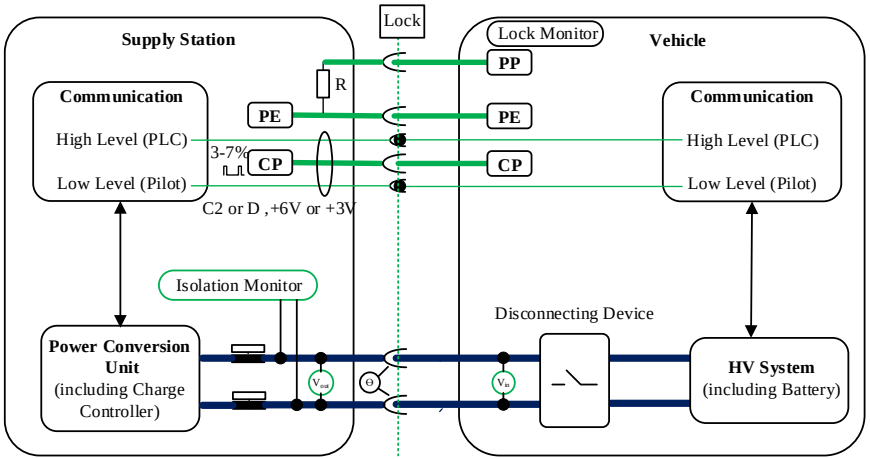
→ Maximum current to be allowed during pre-charge

→ Required DC voltage

- Start of pre-charge stage
- EV sends Pre-Charge Request, which contains both requested DC current < 2A (maximum inrush current) and requested DC voltage.

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DC CCS2 Charging Sequence Cont.



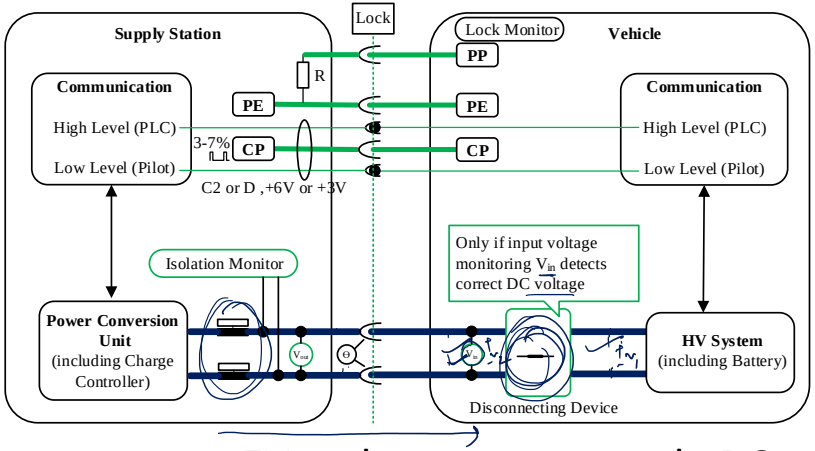
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

Pre-charge from t_5 -t_6

- DC supply adapts DC output voltage within tolerances and limits current to maximum value of 2 A.
- At t_6, DC output voltage reaches requested voltage within tolerances

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DC CCS2 Charging Sequence Cont.



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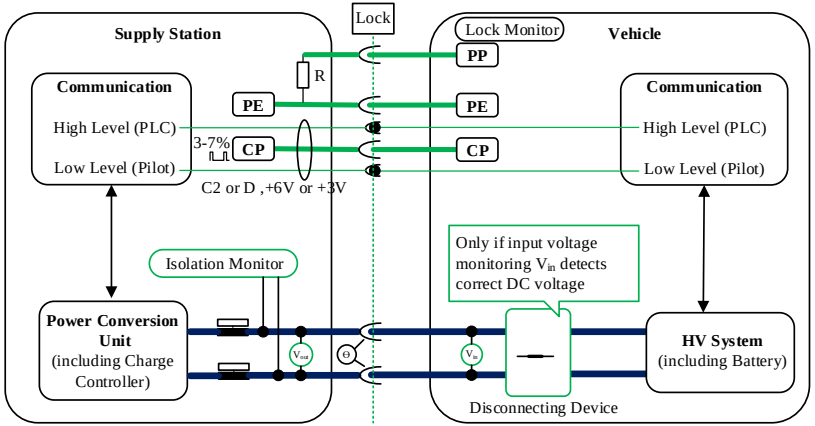
Pre-charge from t_6–t_7

$V_1 - V_2 < 20V$

- If necessary EV adapts requested DC voltage with cyclic messages in order to limit deviation of DC output voltage from EV battery voltage to less than 20 V.
- At t_7, EV closes disconnecting device after deviation of DC output voltage from EV battery voltage is less than 20 V.

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DC CCS2 Charging Sequence Cont.



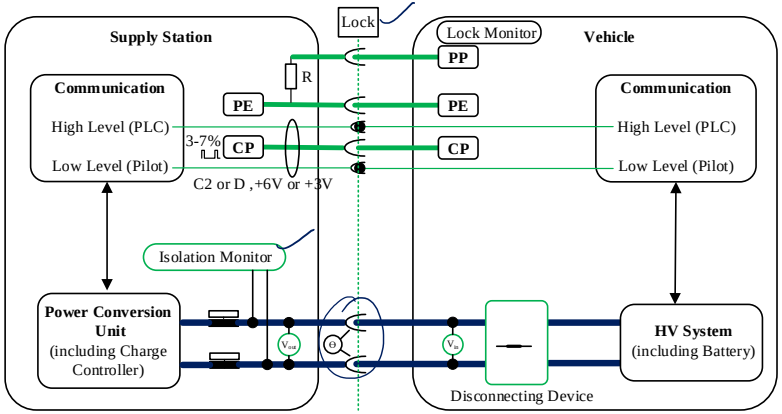
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

Pre-charge from t_7–t_8

- EV sends Power Delivery Request to enable DC power supply output.
- After DC supply gives feedback that it is ready for energy transfer EV sets DC current request to start energy transfer phase at t_8.

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DC CCS2 Charging Sequence Cont.



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Charging stage

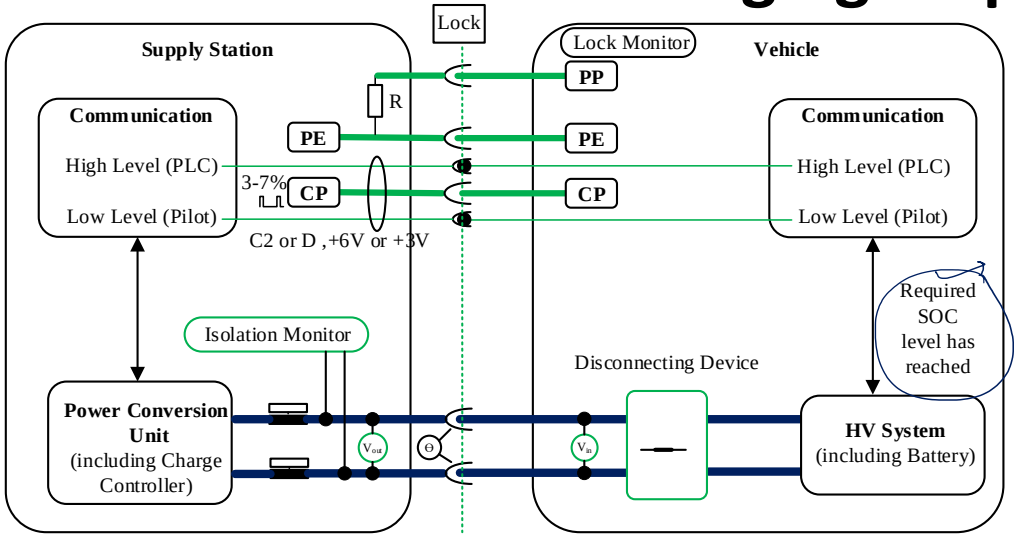
- From $t_8 - t_9$, DC supply adapts its output current and voltage to the requested values.
- DC supply responds with output current and voltage, its current limit and voltage limit, and its present status.

$T_d = t_9 - t_8$

- At t_9, DC output current reaches DC current request with a delay time.
- From t_9 onwards, EV is initiating message cycles by requesting voltage/current. Supply is responding with voltage/current adjustment as well as present limit and status values
- Continuous monitoring of lock, isolation, voltage, current and temperature.

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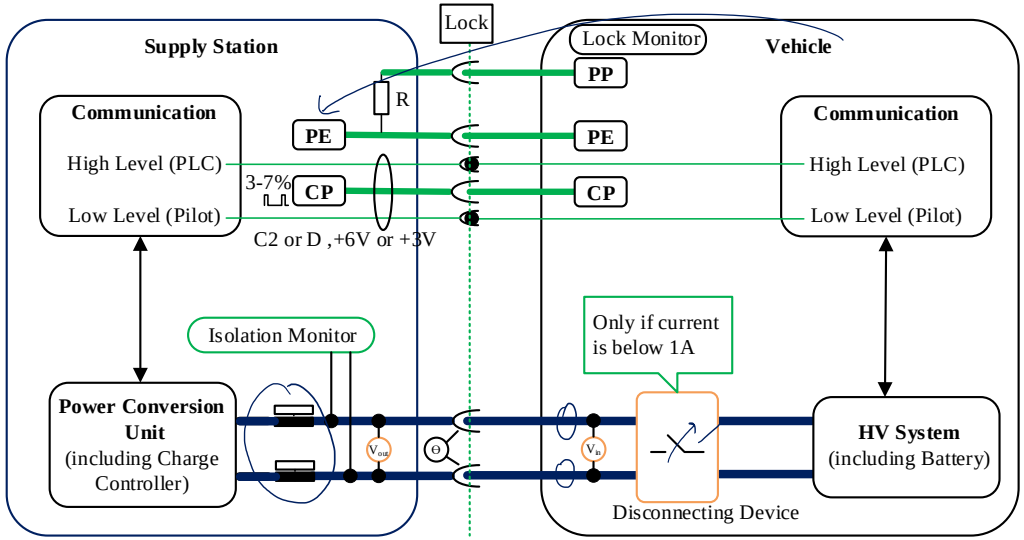
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

Charging stage from t_{10}–t_{11}

- EV reduces the current request to complete the energy transfer.
- The DC supply follows the current request with a time delay and reduces the output current to less than 1A before disabling its output.

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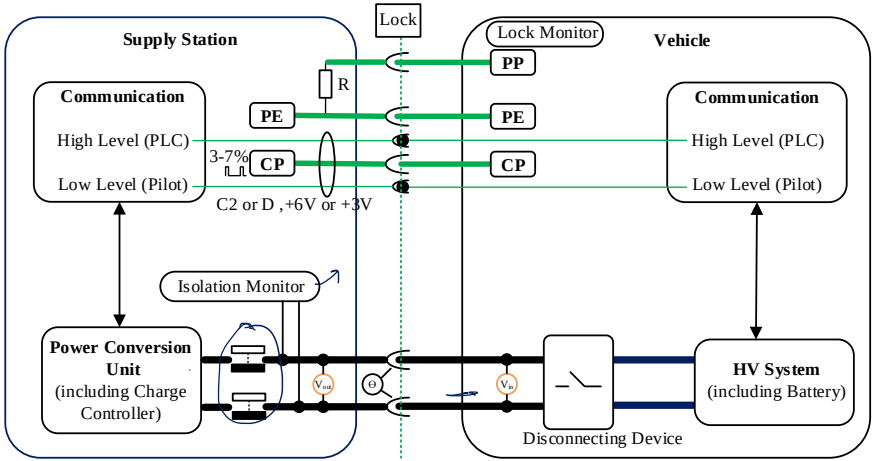
Charging sequence is given in IEC 61851-23 and also being included in AIS-138 part 2

Power down stage from $t_{11} - t_{12}$

- EV sends a message to DC supply to disable its power output
- After current is below 1 A the EV opens its disconnection device

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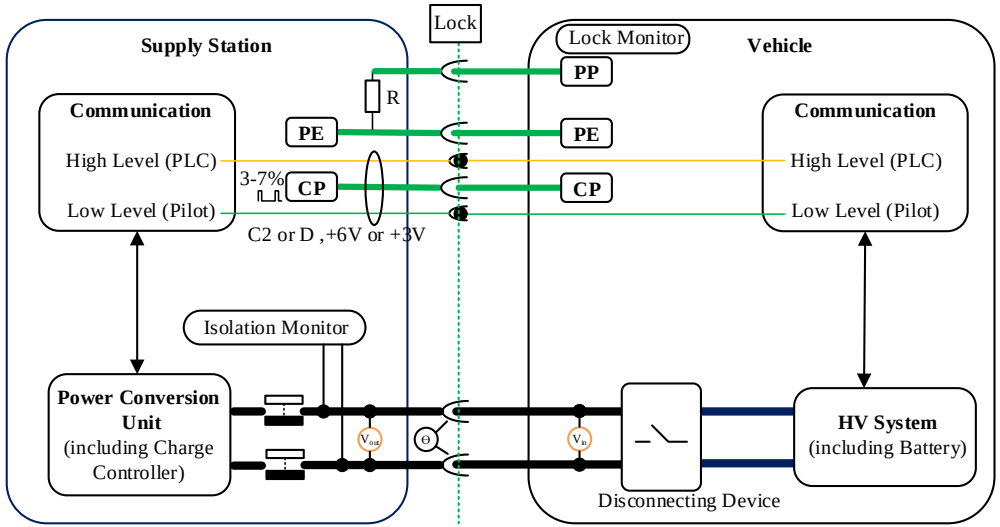
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Power down stage

- At t_{12}, DC supply disables its output and opens contactors.
- DC supply shall not cause any current flow on EV input during discharge

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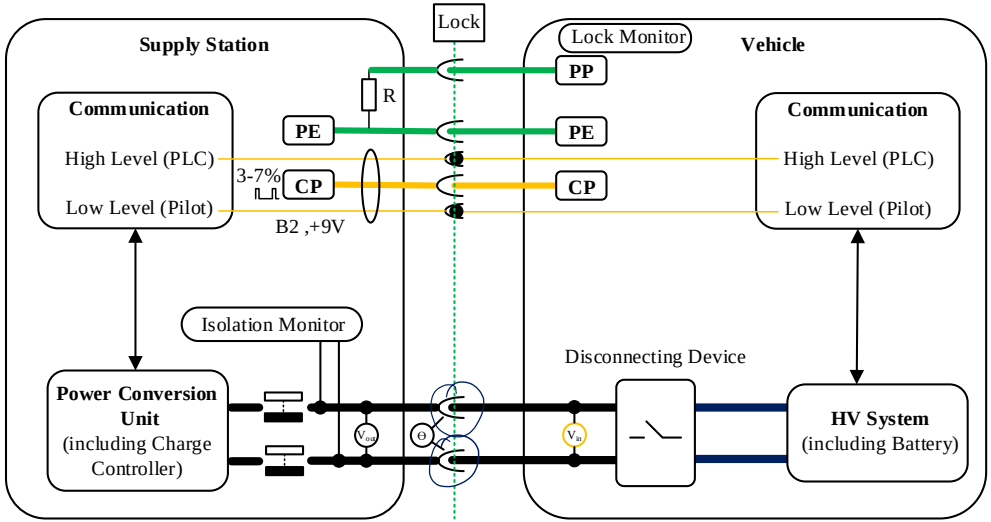
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Power down stage

- At t_{13}, DC supply reports status code ”Not Ready” with a message to indicate it has disabled its output

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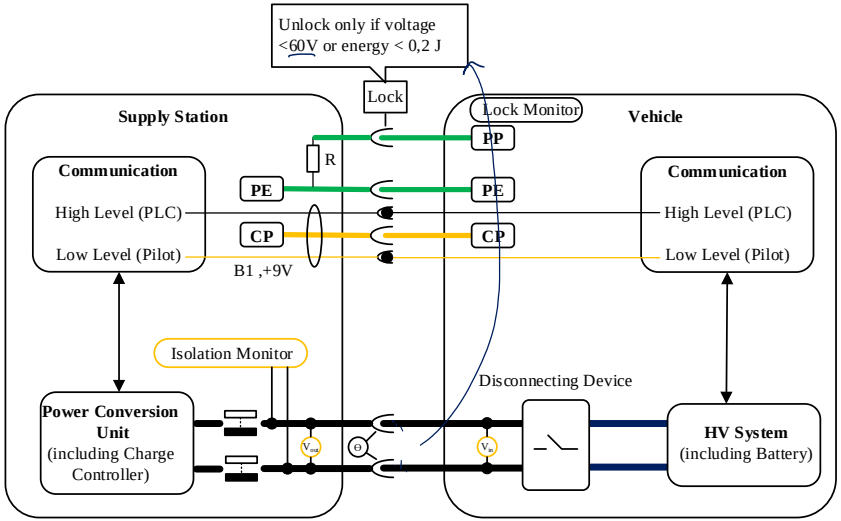
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Power down stage from t_{14}-t_{15}

- EV changes CP state to B after receiving message or after timeout.
- Vehicle may perform welded contactor check and indicate this to DC supply

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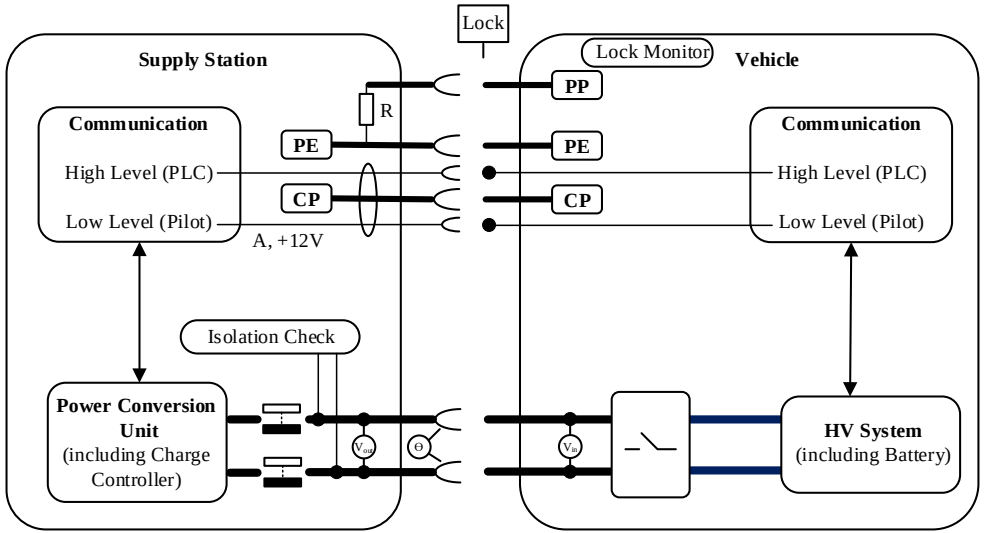
Power down stage

- Session Stop Request with a message and terminates digital communication (PLC).

- From $t_{15} - t_{16}$, EV can start EV isolation monitoring.
- At t_{16}, EV unlocks the connector after DC output has dropped below 60 V
- The DC supply continues isolation monitoring dependent on DC supply strategy.

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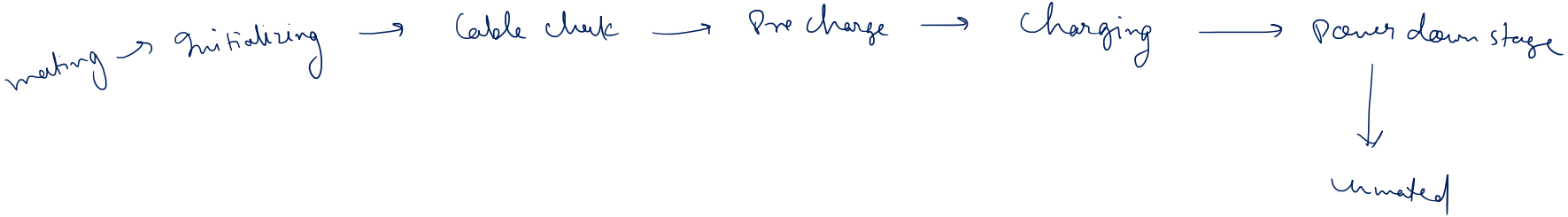
Unmated stage

- Disconnecting of vehicle connector changes CP state from B to A
- PLC is terminated.

- At t_{17}, EV and Supply unmated.
- Supply disables DC output.
- Lock is disabled.

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Thank You